

QAZVIN, Iran, Islamic Republic ofWMO: **407310**Lat: **36.240N**Lon: **50.047E**Elev: **4184**StdP: **12.61**Time Zone: **3.50 (IRN)**Period: **95-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.0	20.6	6.5	9.0	25.3	10.1	10.8	30.4	20.9	46.9	18.3	44.8	0.9	30	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	30.0	100.2	63.7	97.1	63.2	94.9	62.7	67.4	90.8	66.2	89.0	65.1	87.7	9.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.5	88.8	75.8	58.4	85.3	75.4	57.1	81.3	74.7	34.4	91.0	33.3	89.4	32.3	87.6	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	18.5	14.9	12.1	DB	10.7	104.5	6.5	2.8	6.0	106.5	2.2	108.1	-1.4	109.7	-6.2	111.7
(5)				WB	9.9	70.8	6.0	1.8	5.6	72.1	2.1	73.1	-1.3	74.1	-5.6	75.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.4	35.4	39.3	48.3	55.9	65.8	76.0	80.8	79.3	72.1	60.9	46.8	39.3
	DBStd	16.90	7.85	7.24	7.34	6.42	5.14	4.60	4.08	3.80	4.93	6.04	6.76	5.76
	HDD50	1369	453	303	118	21	0	0	0	0	0	3	137	334
	HDD65	3994	918	720	520	281	53	0	0	0	6	152	547	797
	CDD50	4447	0	4	65	197	490	781	956	907	662	342	40	3
	CDD65	1594	0	0	1	7	78	331	491	442	218	26	0	0
	CDH74	20568	0	0	20	110	1135	4288	6363	5564	2660	427	1	0
	CDH80	10841	0	0	1	19	333	2272	3786	3167	1181	82	0	0
Wind	WSAvg	4.0	3.4	4.1	4.7	4.4	4.2	4.6	4.7	4.2	4.1	3.5	3.0	3.3
Precipitation	PrecAvg	13.2	1.4	1.6	2.1	2.2	1.3	0.2	0.2	0.1	0.1	0.9	1.5	1.7
	PrecMax	20.0	2.8	3.1	5.0	4.9	4.6	1.2	0.9	1.8	0.5	3.6	7.6	4.0
	PrecMin	7.8	0.2	0.5	0.3	0.7	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.2
	PrecStd	2.9	0.7	0.8	1.0	1.2	1.0	0.3	0.2	0.4	0.1	0.9	1.5	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.2	62.8	76.8	82.6	90.3	100.5	104.1	101.9	95.2	86.0	71.2	62.2
		MCWB	43.4	46.5	52.0	56.5	59.9	62.5	64.9	63.3	61.4	56.9	50.5	46.1
	2%	DB	53.7	58.5	70.0	76.9	86.3	96.8	100.8	98.6	92.7	82.3	66.4	55.8
		MCWB	41.7	43.6	48.9	54.8	58.5	62.0	64.0	63.0	60.9	55.7	48.9	44.1
	5%	DB	50.3	55.1	66.0	73.0	83.8	93.9	98.7	96.3	89.7	79.0	62.8	52.2
		MCWB	40.1	42.1	47.5	53.4	57.9	61.5	63.9	62.9	59.8	54.7	47.8	42.0
	10%	DB	47.0	51.7	62.2	69.4	80.6	91.4	95.4	93.6	87.4	75.3	59.2	49.6
		MCWB	38.8	40.8	46.1	52.0	57.8	61.0	63.6	62.5	59.3	53.9	46.3	41.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.2	47.7	53.3	58.8	63.5	66.6	69.7	68.6	65.9	59.5	54.1	48.1
		MCDB	52.7	58.3	70.9	74.2	81.8	89.9	93.8	92.3	86.7	80.4	64.0	57.4
	2%	WB	43.5	45.5	50.6	56.6	61.4	65.0	67.9	67.1	63.7	57.5	51.5	45.7
		MCDB	50.2	54.4	66.2	70.7	79.9	88.6	92.0	90.0	85.4	76.5	60.7	53.8
	5%	WB	41.4	43.6	48.9	55.1	60.0	63.8	66.7	65.8	62.2	56.2	49.6	43.7
		MCDB	48.4	51.7	62.4	68.8	78.2	87.1	90.0	88.0	83.6	73.8	59.2	50.4
	10%	WB	39.4	41.8	47.2	53.7	58.7	62.7	65.6	64.6	60.9	54.8	47.8	42.0
		MCDB	46.3	49.8	59.2	65.8	76.4	85.5	88.6	86.8	82.1	71.3	57.2	47.6
Mean Daily Temperature Range	5% DB	MDBR	18.6	19.7	22.4	23.4	27.0	30.3	30.0	29.9	29.7	25.5	19.9	18.0
		MCDBR	21.9	24.6	27.0	28.0	31.2	32.4	32.4	31.6	31.2	29.8	24.9	22.4
		MCWBR	12.9	13.1	12.8	12.1	11.5	9.9	9.6	10.2	11.5	12.3	12.2	13.0
	5% WB	MCDBR	19.2	20.4	23.3	24.2	28.5	31.0	30.5	30.1	30.2	26.9	20.7	19.4
		MCWBR	12.1	11.9	11.8	11.8	11.5	10.4	9.9	9.9	11.5	11.6	10.6	12.0
Clear-Sky Solar Irradiance	taub	0.315	0.351	0.406	0.479	0.476	0.468	0.509	0.463	0.411	0.400	0.347	0.315	
	taud	2.321	2.212	2.072	1.924	1.968	2.003	1.920	2.039	2.164	2.154	2.279	2.351	
	Ebn at Noon	277	278	271	257	259	260	249	258	266	256	261	268	
	Edh at Noon	30	38	49	60	58	56	60	52	44	40	32	28	
All-Sky Solar Radiation	RadAvg	789	1024	1327	1646	2063	2415	2342	2174	1797	1278	846	688	
	RadStd	68	107	120	174	120	112	78	93	65	90	106	94	

Historical Trends

	DBAvg	Heating	
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